HW 2 - Redlining and Environmental Injustice

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Part 1: Create a map of historical redlining neighborhoods

```
# Load packages and data

library(sf)
library(tmap)
library(tidyverse)
library(spData)
library(knitr)
library(gt)
set_theme(theme_bw())


ej_screen <- st_read(
  'data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  )
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/michaela/github/eds223/EDS223-HW2/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_C
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
redline <- st_read(
  'data/mapping-inequality/mapping-inequality-los-angeles.json'
  )
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/Users/michaela/github/eds223/EDS223-HW2/data/mapping-inequality/mapping-
inequality-los-angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```
crashes <- st_read(
  'data/SWITRS_LA/Crashes.csv'
  )
```

```
Reading layer `Crashes' from data source
  `/Users/michaela/github/eds223/EDS223-HW2/data/SWITRS_LA/Crashes.csv'
  using driver `CSV'
```

```
# filter to the county we are interested in (LA)
la <- ej_screen %>%
  dplyr::filter(CNTY_NAME %in% c("Los Angeles County"))
```

```
# Clean bicycle and pedestrian crash data

crashes_clean <- crashes |>
    mutate(
      POINT_X = as.numeric(
        POINT_X), # started as characters, so we need to make them numeric
           POINT_Y = as.numeric(
            POINT_Y)) |>
    filter(
      !is.na(POINT_X),
       !is.na(POINT_Y)) # take out the nas
```

This dataset has latitude and longitude columns in addition to `POINT_X` and `POINT_Y` columns. By reading the metadata found here, we can discern that `POINT_X` and `POINT_Y` are the better choices based on amount of NAs and the precision of data.

```
# Make crashes data into a spatial object

crashes_sf <- st_as_sf(crashes_clean,
```

```
                  coords = c(
                    "POINT_X", "POINT_Y"),
                  crs = 4326)    # WGS84 -- from metadata


# create a warning to test if the crs's match
if(st_crs(crashes_sf) != st_crs(redline)){
  warning("coordinate reference systems do not match")
} else {
  print("match!")
}
```

```
[1] "match!"
```

```
# Check the other objects' crs.

if(st_crs(la) != st_crs(redline)){
  warning("coordinate reference systems do not match")
  } else {
  print("match!")
}
```

```
# Transform the objects so all have the same crs

redline <- st_transform(redline, 3857)
if(st_crs(la) != st_crs(redline)){
  warning("coordinate reference systems do not match")
  } else {
  print("match!")
}
```

```
[1] "match!"
```

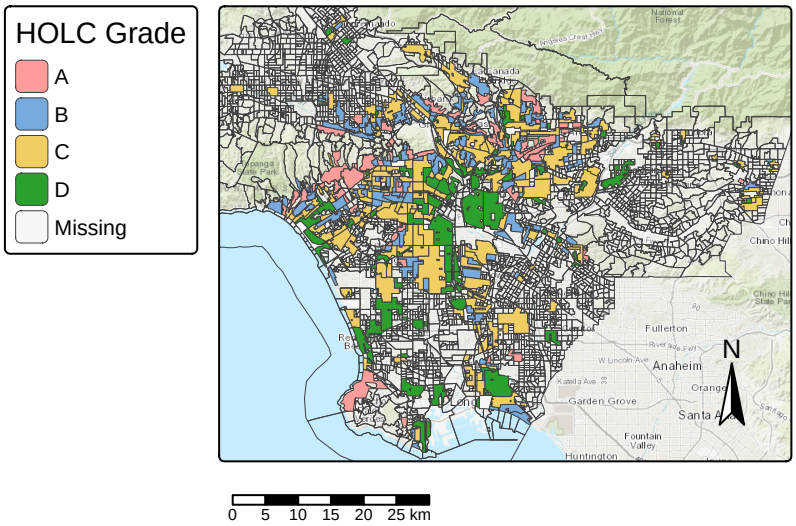
```
crashes_sf <- st_transform(crashes_sf, 3857)
if(st_crs(crashes_sf) != st_crs(redline)){
  warning("coordinate reference systems do not match")
} else {
  print("match!")
}
```

```
[1] "match!"
```

Map HOLC grade by neighborhood

```
tm_basemap(server = "Esri.WorldTopoMap") + # choose basemap
tm_shape(la, bbox = redline) + # set the bbox to where there is info on redlining
    tm_borders(lwd = 0.3, ) +
tm_shape(redline) +
    tm_polygons(fill = 'grade', lwd = 0.1, # color by HOLC grade
    fill.legend = tm_legend("HOLC Grade")) +
tm_scalebar(position = tm_pos_out("center", "bottom")) +
tm_title("HOLC Grade by Neighborhood in Los Angeles County") +
tm_legend(position = tm_pos_out("left", "center")) +
tm_compass(position = tm_pos_in(
      "right",
       "bottom"))
```

HOLC Grade by Neighborhood in Los Angeles County




```
# Create a table summarizing the percentage of
# census block groups that fall within each HOLC grade

# Join using within
holc_bg <- st_join(la, redline, join = st_within)

# group by the grading and  add a column with the percentage in each grade
```

Census Block Groups by HOLC Grade

Los Angeles County; block groups fully within a HOLC polygon

HOLC Grade	Block groups	Percent of block groups (%)
A	15	0.2275831
B	78	1.1834320
C	509	7.7226521
D	287	4.3544227
No HOLC grade	5702	86.5119102

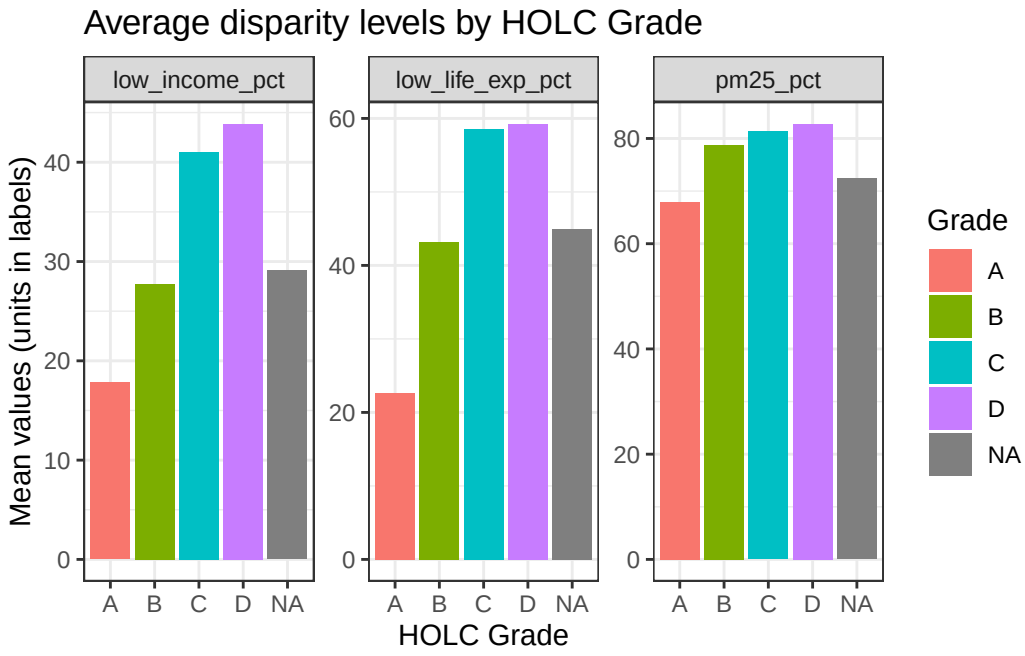
```
holc_bg |>
  st_drop_geometry() |>
  mutate(grade = replace_na(grade, "No HOLC grade")) |>
  count(grade) |>
  mutate(percent = n / nrow(holc_bg) * 100) |>
  gt() |>
  tab_header(title = "Census Block Groups by HOLC Grade",
             subtitle = "Los Angeles County;
             block groups fully within a HOLC polygon") |>
  cols_label(grade = "HOLC Grade",
             n = "Block groups",
             percent = "Percent of block groups (%)")
```

```
# Summarize current conditions (from the EJScreen data) within HOLC grades
# group by means
holc_means <- holc_bg |>
  st_drop_geometry() |>
  group_by(grade) |>
  summarise(low_income_pct = (mean(LOWINCPCT, na.rm = TRUE))*100,
            pm25_pct = mean(P_PM25, na.rm = TRUE),
            low_life_exp_pct = mean(P_LIFEEXPPCT, na.rm = TRUE))
```

```
# plot by variable

# pivot longer
holc_long <- holc_means |>
  pivot_longer(cols = c(low_income_pct, pm25_pct, low_life_exp_pct),
               names_to = "Injustices",
               values_to = "Mean_disparity")
```

```r
# plot and use facet wrap to show all 3
ggplot(holc_long, aes(x = grade, y = Mean_disparity, fill = grade)) +
  geom_col() +
   facet_wrap(~ Injustices, scales = "free_y") +
    labs(title = "Average disparity levels by HOLC Grade",
         x = "HOLC Grade",
         y = "Mean values (units in labels)",
         fill = "Grade")
```



Interpretation

In the graph above we can see that for both percentage of low income households and percentage of low life expectancy that as HOLC grade gets worse (closer to 'D' rating), the average level of each variable increases. Neighborhoods with higher pedestrian fatalities are also experiencing more environmental and health challenges. This pattern suggest a potential relationship that could be explained by historical redlining practices. People residing in low-income neighborhoods may not be receiving the same government attention and funding for proper infrastructure as their high-income counterparts.

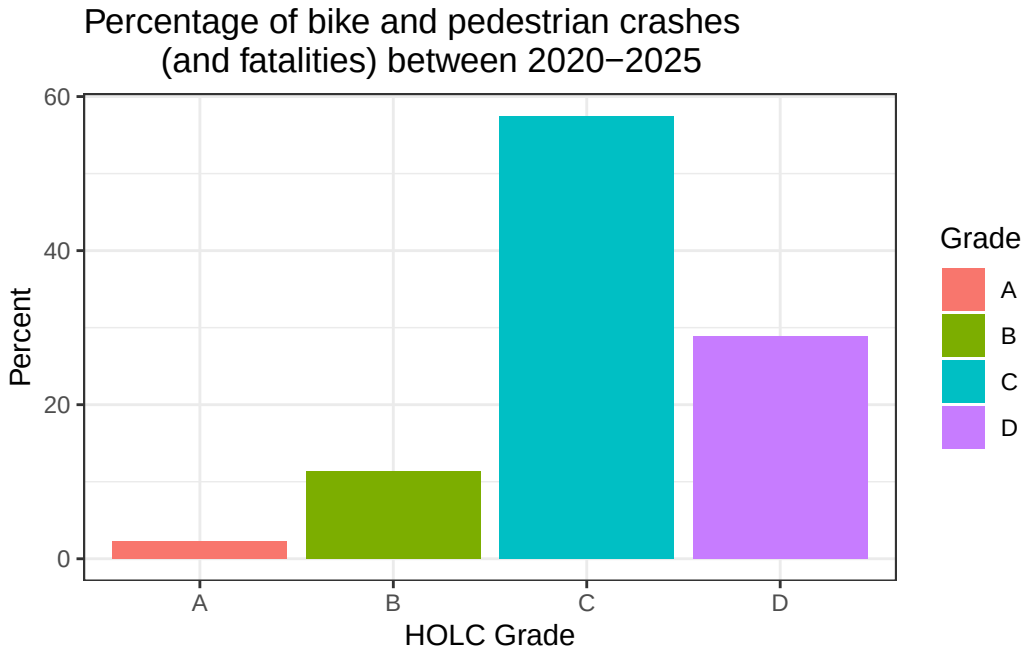
Part 2

Summarizing the percent of observations within redlined neighborhoods within each HOLC grade

```
# create a new joined data frame with crash data and redlined data
bikeped <- crashes_sf |>
  st_filter(la) |>       # filter to just la
  st_join(redline, join = st_within)

# create a summary data fram with the percentage of accidents
bikeped_summary <- bikeped |>
  st_drop_geometry() |>
  mutate(ACCIDENT_YEAR = as.numeric(ACCIDENT_YEAR)) |>
  filter(ACCIDENT_YEAR >= 2020 & ACCIDENT_YEAR <=2025) |>
  filter(BICYCLE_ACCIDENT == "Y" | PEDESTRIAN_ACCIDENT == "Y") |>
  drop_na(grade) |>
  group_by(grade) |>
  summarise(crashes = n()) |>
  mutate(percent = ((crashes)/sum(crashes))*100)
```

```
ggplot(bikeped_summary, aes(x = grade, y = percent, fill = grade)) +
    geom_col() +
    labs(
      title =
        "Percentage of bike and pedestrian crashes
        (and fatalities) between 2020-2025",
        x = "HOLC Grade",
        y = "Percent",
        fill = "Grade")
```



References:

"TIMS - Transportation Injury Mapping System." Berkeley.Edu, 2017, https://tims.berkeley.edu/help/crashdata/switrs-codebook.php. Accessed 23 Oct. 2026.